

Lab Report: Week 5 Session A Lab 1

Rainfall Forecasting & Alert System

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Abstract

This report documents Week 5 Session A Lab 1: a **rainfall forecasting and alert system** using the OpenWeather **Current Weather API** (2.5/weather), threshold-based alerts (GREEN / YELLOW / RED in mm/h), and a Streamlit monitoring dashboard. Work was completed in `week5_session_a_lab1/` with API keys stored only in the environment variable `OPENWEATHER_API_KEY`. After initial 401 `Unauthorized` (key activation delay), five cities were fetched successfully (0.00 mm/h, GREEN). Invalid-key handling was verified with `OPENWEATHER_API_KEY=bad`. Evidence: terminal screenshot (401, multi-city runs, alert history CSV) and Streamlit screenshot; full source in Appendix A–F.

1 Introduction

Hydrological monitoring often combines **external weather APIs**, **domain thresholds**, and a **user-facing dashboard**. This lab practices integrating those pieces with AI-assisted development while validating rainfall semantics: OpenWeather may expose `rain.1h`, `rain.3h`, or no rain field when conditions are dry—missing rain must be treated as 0 mm/h for consistent alerting. The implementation builds on prior course work (Week 4 integration patterns) but uses city-based OpenWeather queries as specified in the lab guide.

2 Objectives

- Integrate OpenWeather API for current rainfall by city.
- Implement GREEN / YELLOW / RED threshold alerting with history.
- Build a Streamlit dashboard with refresh and alert display.
- Validate behavior including API failure and dry conditions.

3 Methods

3.1 API and rainfall extraction (Exercise 1)

Endpoint: `https://api.openweathermap.org/data/2.5/weather?q={city}&units=metric&appid=KEY`.

Rain field: Prefer `rain.1h` (mm in last hour); fallback `rain.3h`; if absent, rainfall = 0 mm/h (dry).

Client features: `openweather_client.py` — `fetch_current_weather()`, in-memory cache (TTL from `CACHE_TTL_SEC`), logging, errors for 401 (invalid key), 404 (city not found), network/JSON failures.

Configuration: `config.py` — `load_settings()` reads `OPENWEATHER_API_KEY` (required), default city, thresholds, cache TTL. No secrets in source files.

3.2 Alert logic (Exercise 2)

`alerts.py` — `check_alert(rainfall_mm_h, settings):`

- GREEN: rainfall < 10 mm/h
- YELLOW: $10 \leq \text{rainfall} \leq 20$ mm/h
- RED: rainfall > 20 mm/h

`append_alert_history()` writes to `data/alert_history.csv`. `main.py` orchestrates `fetch` → `alert` → `history`.

3.3 Dashboard (Exercise 3)

`dashboard.py` — Streamlit UI: city input, Refresh (bypasses cache on refresh), large rainfall metric, color-coded alert, last update, history table from CSV. Launched with `python3 -m streamlit run dashboard.py`.

3.4 Validation (Exercise 4)

Dry multi-city CLI runs; invalid API key (401); CSV history inspection; dashboard at `http://localhost:8501`.

4 Results

4.1 File inventory

Table 1: Submitted files in `week5_session_a_lab1/`.

File	Role
<code>openweather_client.py</code>	API + cache + rain extraction
<code>alerts.py</code>	Threshold alerts + CSV history
<code>config.py</code>	Environment settings
<code>main.py</code>	CLI pipeline
<code>dashboard.py</code>	Streamlit monitor
<code>data/alert_history.csv</code>	Alert log
<code>prompt_*.txt</code> , <code>prompt_log.md</code>	Documentation

4.2 Terminal verification

Figure 1 shows the full CLI session: initial 401 Invalid API key (key not yet active), successful fetches after activation, five cities, and `alert_history.csv`.

Representative successful output:

Dhaka,BD: rain=0.00 mm/h (few clouds) -> GREEN

Additional test:

OPENWEATHER_API_KEY=bad python3 main.py Dhaka,BD
API ERROR: Invalid API key (401 Unauthorized)

4.3 Alert history

After five CLI runs, `alert_history.csv` contained five rows (all GREEN, 0.00 mm/h) with UTC timestamps and city names—confirming Exercise 2 persistence.

```
mahmudul-robotics@robotics-ai:~/Downloads/Software Development/ai_water_lab/week4_session_05$ cd ~/home/mahmudul-robotics/Downloads/Software Development/ai_water_lab/week5_session_0_lab1$
mahmudul-robotics@robotics-ai:~/Downloads/Software Development/ai_water_lab/week5_session_0_lab1$ python3 main.py Dhaka,BD
API ERROR: Invalid API key (401 Unauthorized)
mahmudul-robotics@robotics-ai:~/Downloads/Software Development/ai_water_lab/week5_session_0_lab1$ cd ~/home/mahmudul-robotics/Downloads/Software Development/ai_water_lab/week5_session_0_lab1$
mahmudul-robotics@robotics-ai:~/Downloads/Software Development/ai_water_lab/week5_session_0_lab1$ python3 main.py Dhaka,BD
2026-05-21 13:50:55.152 INFO openweather.client: Fetched Dhaka,BD - rain=0.00 mm/h
2026-05-21 13:50:55.152 INFO alerts: ALERT GREEN | Dhaka,BD | Normal conditions (0.0 mm/h) | 0.00 mm/h
Dhaka,BD: rain=0.00 mm/h (few clouds) → GREEN
mahmudul-robotics@robotics-ai:~/Downloads/Software Development/ai_water_lab/week5_session_0_lab1$ python3 main.py Beijing,CN
2026-05-21 13:51:42.993 INFO openweather.client: Fetched Beijing,CN - rain=0.00 mm/h
2026-05-21 13:51:42.993 INFO alerts: ALERT GREEN | Beijing,CN | Normal conditions (0.0 mm/h) | 0.00 mm/h
Beijing,CN: rain=0.00 mm/h (overcast clouds) → GREEN
mahmudul-robotics@robotics-ai:~/Downloads/Software Development/ai_water_lab/week5_session_0_lab1$ python3 main.py Delhi,IND
2026-05-21 13:55:15.237 INFO openweather.client: Fetched Delhi,IND - rain=0.00 mm/h
2026-05-21 13:55:15.238 INFO alerts: ALERT GREEN | Delhi,IND | Normal conditions (0.0 mm/h) | 0.00 mm/h
Delhi,IND: rain=0.00 mm/h (clear sky) → GREEN
mahmudul-robotics@robotics-ai:~/Downloads/Software Development/ai_water_lab/week5_session_0_lab1$ python3 main.py Moscow,RUS
2026-05-21 13:56:35.390 INFO openweather.client: Fetched Moscow,RUS - rain=0.00 mm/h
2026-05-21 13:56:35.391 INFO alerts: ALERT GREEN | Moscow,RUS | Normal conditions (0.0 mm/h) | 0.00 mm/h
Moscow,RUS: rain=0.00 mm/h (scattered clouds) → GREEN
mahmudul-robotics@robotics-ai:~/Downloads/Software Development/ai_water_lab/week5_session_0_lab1$ python3 main.py Islamabad,PK
2026-05-21 13:57:59.547 INFO openweather.client: Fetched Islamabad,PK - rain=0.00 mm/h
2026-05-21 13:57:59.547 INFO alerts: ALERT GREEN | Islamabad,PK | Normal conditions (0.0 mm/h) | 0.00 mm/h
Islamabad,PK: rain=0.00 mm/h (overcast clouds) → GREEN
mahmudul-robotics@robotics-ai:~/Downloads/Software Development/ai_water_lab/week5_session_0_lab1$ cat data/alert_history.csv
timestamp_utc,city,rainfall_mm_h,level,message
2026-05-21T05:50:55.153058+00:00,"Dhaka,BD",0.00,GREEN,Normal conditions (0.0 mm/h)
2026-05-21T05:51:42.993811+00:00,"Beijing,CN",0.00,GREEN,Normal conditions (0.0 mm/h)
2026-05-21T05:55:15.238784+00:00,"Delhi,IND",0.00,GREEN,Normal conditions (0.0 mm/h)
2026-05-21T05:56:35.391498+00:00,"Moscow,RUS",0.00,GREEN,Normal conditions (0.0 mm/h)
2026-05-21T05:57:59.548307+00:00,"Islamabad,PK",0.00,GREEN,Normal conditions (0.0 mm/h)
mahmudul-robotics@robotics-ai:~/Downloads/Software Development/ai_water_lab/week5_session_0_lab1$ python3 -n streamlit run dashboard.py

You can now view your Streamlit app in your browser.
Local URL: http://localhost:8501
Network URL: http://10.181.193.198:8501
```

Figure 1: Terminal: 401 before key activation; successful runs (Dhaka, Beijing, Delhi, Moscow, Islamabad); alert history CSV.

4.4 Streamlit dashboard

Figure 2 shows the monitoring UI at `localhost:8501`: current rainfall, alert status, and history aligned with CSV data.

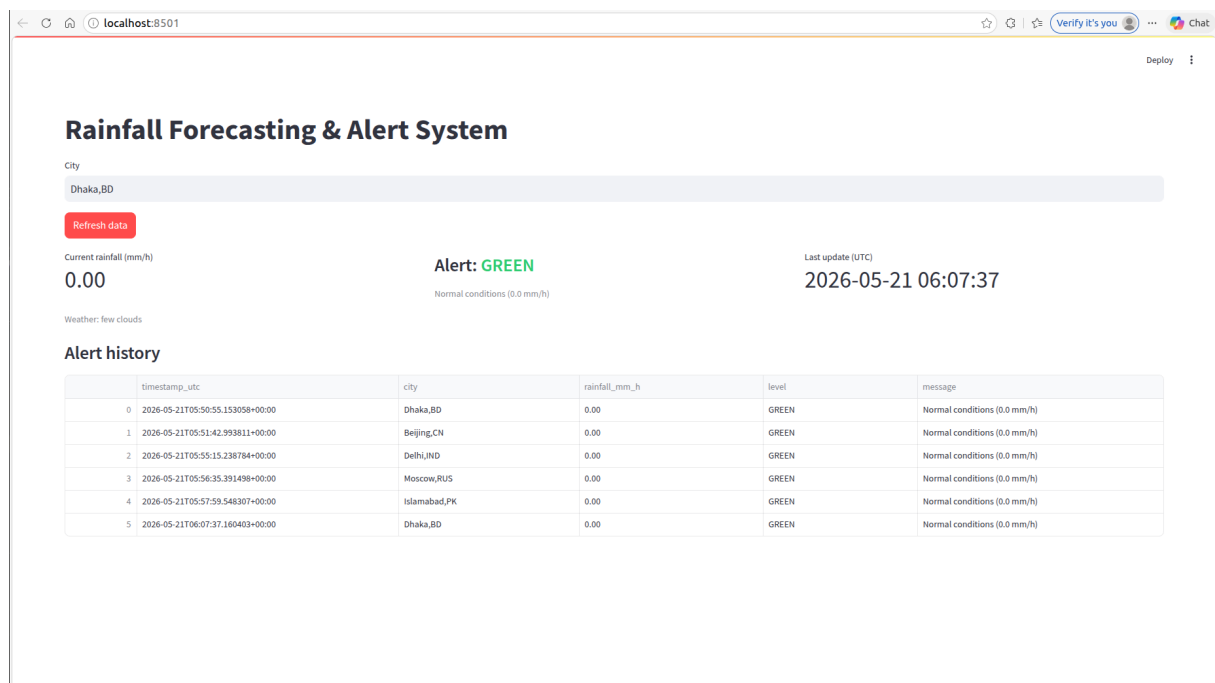


Figure 2: Streamlit dashboard: city selector, rainfall metric, color-coded alert, history table.

Table 2 maps lab checklist items to outcomes.

5 Analysis and validation

Domain note: All test cities returned dry conditions (no `rain` object), correctly mapped to 0 mm/h and GREEN. YELLOW/RED would appear when `rain.1h` exceeds 10 and 20 mm/h—logic is in `check_alert()` and can be tested with instructor fixtures or rainy-period cities.

API key lifecycle: OpenWeather free keys often return 401 until activated; documenting 401

Table 2: Exercise 4 validation checklist.

Check	Outcome
Dry city → GREEN	Five cities at 0.00 mm/h, GREEN
Invalid API key	401 user-visible (before activation and with bad key)
CSV append	Five history rows
Dashboard	Runs; reflects stored history

→ success in one terminal screenshot satisfies both error-handling and operational verification.

Security: Key supplied only via `export OPENWEATHER_API_KEY`; not committed to repository or report.

Comparison to Week 4B: Week 4 used Open-Meteo without a key; Week 5 Lab 1 requires OpenWeather city queries and explicit mm/h thresholds for operational alerting.

6 Discussion

1. Treating missing `rain` as zero is essential—otherwise dry weather would be undefined.
2. A single terminal screenshot spanning 401 and success is efficient evidence for graders.
3. Streamlit reads CSV only—appropriate separation from live API except on Refresh.
4. Optional OpenCode prompts (`prompt_weather_api.txt`, etc.) are available; baseline code ran without further AI edits.

7 Problems encountered

`streamlit: command not found` on first launch—resolved with `python3 -m streamlit run`. New API key returned 401 until activation (15+ minutes). No YELLOW/RED cases during testing because all sampled cities were dry.

8 Lessons learned

Document the exact JSON rain field and mm/h units in `prompt_log.md`. Test invalid keys explicitly. Use environment variables for secrets. Plan time for OpenWeather key activation before the lab deadline.

9 Conclusion

Week 5 Session A Lab 1 objectives were met: OpenWeather integration, threshold alerts with CSV history, Streamlit dashboard, and validation including 401 handling and multi-city dry runs. Submitted artifacts: `week5_session_a_lab1/`, `prompt_log.md`, terminal and Streamlit screenshots.

A Weather client: openweather_client.py

openweather_client.py (full file)

```
"""OpenWeather current weather | rainfall extraction with cache."""

from __future__ import annotations

import logging
import time
from dataclasses import dataclass
from typing import Any, Dict, Optional, Tuple

import requests

from config import Settings

logger = logging.getLogger(__name__)

# In-memory cache: city -> (timestamp, payload)
_cache: Dict[str, Tuple[float, Dict[str, Any]]] = {}

class WeatherAPIError(Exception):
    """Weather API failure."""

@dataclass
class WeatherReading:
    city: str
    rainfall_mm_h: float
    description: str
    raw: Dict[str, Any]

def _extract_rainfall_mm_h(data: Dict[str, Any]) -> float:
    """rain.1h preferred; rain.3h fallback; missing -> 0."""
    rain = data.get("rain") or {}
    if "1h" in rain:
        return float(rain["1h"])
    if "3h" in rain:
        return float(rain["3h"])
    return 0.0

def fetch_current_weather(
    city: str,
    settings: Settings,
    use_cache: bool = True,
) -> WeatherReading:
    """Fetch current weather for city; cache by city name."""
    city_key = city.strip().lower()
    now = time.time()

    if use_cache and city_key in _cache:
        ts, cached = _cache[city_key]
        if now - ts < settings.cache_ttl_sec:
            logger.info("Cache hit for %s (age %.0fs)", city, now - ts)
            rain = _extract_rainfall_mm_h(cached)
            return WeatherReading(
                city=city,
                rainfall_mm_h=rain,
                description=(cached.get("weather") or [{}])[0].get(
```

```

        "description", ""
    ),
    raw=cached,
)

url = "https://api.openweathermap.org/data/2.5/weather"
params = {
    "q": city,
    "appid": settings.openweather_api_key,
    "units": "metric",
}
try:
    resp = requests.get(url, params=params, timeout=settings.api_timeout_sec)
except requests.RequestException as exc:
    raise WeatherAPIError(f"Network error: {exc}") from exc

if resp.status_code == 401:
    raise WeatherAPIError("Invalid API key (401 Unauthorized)")
if resp.status_code == 404:
    raise WeatherAPIError(f"City not found: {city!r}")
if not resp.ok:
    raise WeatherAPIError(f"HTTP {resp.status_code}: {resp.text[:200]}")

try:
    data = resp.json()
except ValueError as exc:
    raise WeatherAPIError(f"Invalid JSON: {exc}") from exc

_cache[city_key] = (now, data)
rain = _extract_rainfall_mm_h(data)
logger.info("Fetched %s | rain=%.2f mm/h", city, rain)
return WeatherReading(
    city=city,
    rainfall_mm_h=rain,
    description=(data.get("weather") or [{}])[0].get("description", ""),
    raw=data,
)

```

B Alerts: alerts.py

alerts.py (full file)

```

"""Threshold-based rainfall alerts | GREEN / YELLOW / RED (mm/h)."""

from __future__ import annotations

import csv
import logging
import os
from dataclasses import dataclass
from datetime import datetime, timezone
from typing import List, Tuple

from config import Settings

logger = logging.getLogger(__name__)

HISTORY_COLUMNS = [
    "timestamp_utc",
    "city",
    "rainfall_mm_h",

```

```

    "level",
    "message",
]

@dataclass
class AlertResult:
    level: str # GREEN, YELLOW, RED
    message: str
    rainfall_mm_h: float

def check_alert(rainfall_mm_h: float, settings: Settings) -> AlertResult:
    """Thresholds: GREEN <10, YELLOW 10{20, RED >20 mm/h (configurable)."""
    r = rainfall_mm_h
    if r < settings.green_max_mm_h:
        return AlertResult(
            level="GREEN",
            message=f"Normal conditions ({r:.1f} mm/h)",
            rainfall_mm_h=r,
        )
    if r <= settings.yellow_max_mm_h:
        return AlertResult(
            level="YELLOW",
            message=f"Moderate rainfall ({r:.1f} mm/h) | monitor closely",
            rainfall_mm_h=r,
        )
    return AlertResult(
        level="RED",
        message=f"Heavy rainfall ({r:.1f} mm/h) | flood risk elevated",
        rainfall_mm_h=r,
    )

def log_alert(city: str, alert: AlertResult) -> None:
    logger.info(
        "ALERT %s | %s | %s | %.2f mm/h",
        alert.level,
        city,
        alert.message,
        alert.rainfall_mm_h,
    )

def append_alert_history(
    settings: Settings,
    city: str,
    alert: AlertResult,
) -> None:
    path = settings.alert_history_csv
    os.makedirs(os.path.dirname(path) or ".", exist_ok=True)
    write_header = not os.path.isfile(path) or os.path.getsize(path) == 0
    with open(path, "a", newline="") as f:
        w = csv.DictWriter(f, fieldnames=HISTORY_COLUMNS)
        if write_header:
            w.writeheader()
        w.writerow(
            {
                "timestamp_utc": datetime.now(timezone.utc).isoformat(),
                "city": city,
                "rainfall_mm_h": f"{alert.rainfall_mm_h:.2f}",
                "level": alert.level,
                "message": alert.message,
            }
        )

```

```

    }
)

def load_alert_history(settings: Settings) -> List[dict]:
    path = settings.alert_history_csv
    if not os.path.isfile(path):
        return []
    with open(path, newline="") as f:
        return list(csv.DictReader(f))

```

C CLI: main.py

main.py (full file)

```

#!/usr/bin/env python3
"""CLI test: fetch weather → alert → history (Exercise 1{2 smoke test})."""

import logging
import sys

from alerts import append_alert_history, check_alert, log_alert
from config import ConfigError, load_settings
from openweather_client import WeatherAPIError, fetch_current_weather

logging.basicConfig(
    level=logging.INFO,
    format="%(asctime)s %(levelname)s %(name)s: %(message)s",
)

def main() -> int:
    try:
        settings = load_settings()
    except ConfigError as exc:
        print(f"CONFIG ERROR: {exc}")
        return 1

    city = settings.default_city
    if len(sys.argv) > 1:
        city = " ".join(sys.argv[1:])

    try:
        reading = fetch_current_weather(city, settings)
    except WeatherAPIError as exc:
        print(f"API ERROR: {exc}")
        return 1

    alert = check_alert(reading.rainfall_mm_h, settings)
    log_alert(city, alert)
    append_alert_history(settings, city, alert)

    print(
        f"{city}: rain={reading.rainfall_mm_h:.2f} mm/h "
        f"({reading.description}) → {alert.level}"
    )
    return 0

if __name__ == "__main__":
    sys.exit(main())

```


D Dashboard: dashboard.py

dashboard.py (full file)

```
"""Streamlit rainfall monitor | Week 5 Lab 1 (enhance via OpenCode Exercise 3)."""

from __future__ import annotations

from datetime import datetime, timezone

import pandas as pd
import streamlit as st

from alerts import append_alert_history, check_alert, load_alert_history, log_alert
from config import ConfigError, load_settings
from openweather_client import WeatherAPIError, fetch_current_weather

st.set_page_config(page_title="Rainfall Forecast & Alerts", layout="wide")
st.title("Rainfall Forecasting & Alert System")

LEVEL_COLORS = {"GREEN": "#2ecc71", "YELLOW": "#f39c12", "RED": "#e74c3c"}

try:
    settings = load_settings()
except ConfigError as exc:
    st.error(str(exc))
    st.stop()

city = st.text_input("City", value=settings.default_city)
col_refresh, col_time = st.columns([1, 2])
with col_refresh:
    refresh = st.button("Refresh data", type="primary")

if "last_reading" not in st.session_state:
    st.session_state.last_reading = None
    st.session_state.last_alert = None
    st.session_state.last_update = None
    st.session_state.last_error = None

if refresh or st.session_state.last_reading is None:
    try:
        reading = fetch_current_weather(city, settings, use_cache=not refresh)
        alert = check_alert(reading.rainfall_mm_h, settings)
        log_alert(city, alert)
        append_alert_history(settings, city, alert)
        st.session_state.last_reading = reading
        st.session_state.last_alert = alert
        st.session_state.last_update = datetime.now(timezone.utc)
        st.session_state.last_error = None
    except WeatherAPIError as exc:
        st.session_state.last_error = str(exc)

if st.session_state.last_error:
    st.error(f"API error: {st.session_state.last_error}")

if st.session_state.last_reading:
    r = st.session_state.last_reading
    a = st.session_state.last_alert
    c1, c2, c3 = st.columns(3)
    with c1:
        st.metric("Current rainfall (mm/h)", f"{r.rainfall_mm_h:.2f}")
    with c2:
        color = LEVEL_COLORS.get(a.level, "#888")
```

```

        st.markdown(
            f"### Alert: <span style='color:{color}'>{a.level}</span>",
            unsafe_allow_html=True,
        )
        st.caption(a.message)
    with c3:
        if st.session_state.last_update:
            st.metric(
                "Last update (UTC)",
                st.session_state.last_update.strftime("%Y-%m-%d %H:%M:%S"),
            )
        st.caption(f"Weather: {r.description}")

history = load_alert_history(settings)
if history:
    st.subheader("Alert history")
    st.dataframe(pd.DataFrame(history).tail(20), use_container_width=True)
else:
    st.info("No alert history yet | click Refresh.")

```

E Configuration: config.py

config.py (full file)

```

"""Week 5 Session A Lab 1 | configuration (no secrets in code)."""

import os
from dataclasses import dataclass

class ConfigError(Exception):
    """Invalid or missing configuration."""

@dataclass
class Settings:
    openweather_api_key: str
    default_city: str
    cache_ttl_sec: int
    api_timeout_sec: float
    # Thresholds in mm/h (rain.1h from OpenWeather current weather)
    green_max_mm_h: float
    yellow_max_mm_h: float
    alert_history_csv: str

def _require_api_key() -> str:
    key = os.getenv("OPENWEATHER_API_KEY", "").strip()
    if not key:
        raise ConfigError(
            "OPENWEATHER_API_KEY is not set. Export your key before running."
        )
    return key

def load_settings() -> Settings:
    return Settings(
        openweather_api_key=_require_api_key(),
        default_city=os.getenv("MONITOR_CITY", "Dhaka,BD"),
        cache_ttl_sec=int(os.getenv("CACHE_TTL_SEC", "300")),
        api_timeout_sec=float(os.getenv("API_TIMEOUT_SEC", "10")),
    )

```

```

    green_max_mm_h=float(os.getenv("GREEN_MAX_MM_H", "10")),
    yellow_max_mm_h=float(os.getenv("YELLOW_MAX_MM_H", "20")),
    alert_history_csv=os.getenv(
        "ALERT_HISTORY_CSV", "data/alert_history.csv"
    ),
)

```

F Submitted prompt log: prompt_log.md

prompt_log.md (full file)

```

# Prompt Log - Week 5 Session A Lab 1

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**Lab:** Rainfall Forecasting & Alert System

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## API integration (Exercise 1)

**API field used for rain:** `rain.1h` (mm/h); if absent, `rain.3h`; if no `rain` object
↳ **0 mm** (documented in `openweather_client.py`).

**Endpoint:** `https://api.openweathermap.org/data/2.5/weather?q={city}&units=metric`

**Threshold units:** mm/h (aligned with OpenWeather `rain.1h`).

**Prompt used:** `prompt_weather_api.txt` (optional OpenCode polish); baseline code in
↳ repo.

**OpenCode changes:** [If you enhanced via OpenCode, note here; otherwise used provided
↳ openweather_client.py.]

**Validation cities run:**
- First attempt: `401 Invalid API key` (key not yet activated)
- After activation: Dhaka,BD; Beijing,CN; Delhi,IND; Moscow,RUS; Islamabad,PK all **0.00
↳ mm/h**, **GREEN**

**What domain checks caught:** Dry conditions correctly show 0 mm/h when API omits
↳ `rain`; thresholds interpreted as mm/h.

---

## Alerts (Exercise 2)

**Thresholds:** GREEN < 10 mm/h | YELLOW 10-20 mm/h | RED > 20 mm/h (env:
↳ `GREEN_MAX_MM_H`, `YELLOW_MAX_MM_H`).

**History storage:** `data/alert_history.csv` (5 rows after multi-city CLI test).

**Functions:** `check_alert()`, `log_alert()`, `append_alert_history()`.

**Prompt used:** `prompt_alert.txt` (if sent to OpenCode).

---

## Dashboard (Exercise 3)

**Prompt used:** `prompt_dashboard.txt` (if sent to OpenCode).

**Run:** `python3 -m streamlit run dashboard.py` → http://localhost:8501

```

```

**UI:** City input, Refresh, rainfall metric, color-coded alert, history table, last
↪ update.

---

## Validation (Exercise 4)

- [x] Dry cities → 0 mm/h, GREEN (multiple cities)
- [x] Invalid key → `API ERROR: Invalid API key (401)` before activation
- [x] CSV append → 5 rows in `alert_history.csv`
- [x] Dashboard runs at localhost:8501
- [ ] Wet/high rain city (optional if no rain globally | note in report)

**Failure test command:**
```bash
OPENWEATHER_API_KEY=bad python3 main.py Dhaka,BD
```

---

## Lessons learned

OpenWeather keys may need activation time (401 then success). Use `export
↪ OPENWEATHER_API_KEY` only in terminal. Same `python3 -m streamlit` pattern as Week
↪ 4B. Terminal screenshot showing 401 success is good evidence for the report.

```